

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Industrial activities are vital for economic development, but they often contribute to environmental pollution, posing significant challenges to ecosystems and human well-being. The release of pollutants such as particulate matter and gases from industrial processes necessitates robust monitoring systems to assess and mitigate their impact on the environment.

The conventional methods of industrial pollution monitoring have limitations in terms of real-time data acquisition, efficiency, and cost-effectiveness. As industries continue to expand, there is an increasing need for advanced monitoring solutions that provide accurate and timely data for effective decision-making and regulatory compliance.

1.2 OBJECTIVES

The primary objective of this project is to develop an integrated system for industrial pollution monitoring using LabVIEW and Arduino. The system aims to address the shortcomings of existing monitoring techniques by leveraging the capabilities of LabVIEW, a powerful graphical programming platform, and Arduino, a low-cost and versatile microcontroller.

Specific objectives include:

Real-time Monitoring: Implementing a real-time monitoring system to capture instantaneous changes in pollutant levels.

Efficient Data Acquisition: Utilizing LabVIEW for seamless communication with Arduino, ensuring efficient and reliable data acquisition.

Comprehensive Pollutant Measurement: Deploying Arduino-based sensor modules to measure a range of pollutants, including particulate matter, gases, and environmental parameters.

User Friendly Interface: Developing a user-friendly interface on LabVIEW for real-time visualization, trend analysis, and alarm notifications based on predefined thresholds.

Cost-effectiveness and Scalability: Ensuring that the proposed solution is economically viable for widespread implementation in various industrial settings.

1.3 SCOPE

This project focuses on the integration of LabVIEW and Arduino for industrial pollution monitoring, with a specific emphasis on:

- Monitoring key pollutants emitted by industrial processes.
- Providing a scalable and customizable solution applicable to diverse industrial environments.
- Offering a cost-effective alternative for industries to adopt and implement.

1.4 SIGNIFICANCE

The integration of LabVIEW and Arduino in industrial pollution monitoring holds immense significance in advancing environmental sustainability and regulatory compliance. This innovative approach combines LabVIEW's powerful data acquisition and analysis capabilities with Arduino's versatility in sensor interfacing, providing a cost-effective and efficient solution. By seamlessly integrating these technologies, real-time monitoring of various pollutants becomes accessible, enabling timely detection and response to environmental hazards. This system enhances precision and reliability in data collection, contributing to more accurate assessments of industrial emissions and their impact on the environment. Additionally, the user-friendly interface of LabVIEW facilitates easy visualization and interpretation of pollution data, aiding decision-makers in implementing proactive measures for pollution control. The implementation of this combined system empowers industries to meet stringent environmental standards, fostering a sustainable approach to production. Moreover, the open-source nature of Arduino promotes collaboration and customization, allowing for flexibility in adapting the monitoring system to diverse industrial settings. As a result, the LabVIEW and Arduino integration plays a pivotal role in promoting transparency, accountability, and environmental stewardship, ultimately contributing to the global effort to mitigate industrial pollution and create a cleaner, healthier planet.

The significance of this project lies in its potential to revolutionize industrial pollution monitoring through the integration of LabVIEW and Arduino. The combined strengths of these technologies offer a comprehensive, cost-effective, and user-friendly solution that can significantly improve the accuracy and efficiency of environmental data analysis.

In the contemporary landscape of industrial growth, the adverse impact of pollution on the

INDUSTRIAL POLLUTION MONITORING USING LABVIEW AND ARDUINO

environment has become a pressing concern. The "Industrial Pollution Monitoring Using LabVIEW and Arduino" project aims to address this challenge by developing an innovative and efficient system for real-time monitoring of industrial emissions. Leveraging the power of LabVIEW, a versatile graphical programming platform, and the flexibility of Arduino, a widely adopted open-source electronics platform, this project integrates cutting-edge technologies to create a robust solution for monitoring and managing industrial pollution.

Traditional methods of industrial pollution monitoring often fall short in providing real-time data, hindered by delays in data acquisition and analysis. Additionally, the high costs associated with sophisticated monitoring technologies limit their widespread adoption, particularly by smaller industries. Recognizing these challenges, this project seeks to pioneer a solution that not only ensures accurate and immediate data but is also economically viable for industries of all scales.

This project not only addresses the immediate need for efficient pollution monitoring but also aligns with the global push towards sustainable and responsible industrial practices. By employing LabVIEW and Arduino, we aim to create a scalable and adaptable solution that can be easily implemented across diverse industrial settings. The synergy of these technologies will contribute to a more transparent and accountable approach to industrial processes, fostering a healthier and environmentally conscious industrial ecosystem. Through this endeavor, we aspire to bridge the gap between technology and environmental stewardship, paving the way for a cleaner and sustainable industrial future.

This project endeavors to contribute a practical and innovative solution to the realm of industrial pollution monitoring, aligning with the evolving demands of modern industries for real-time, accurate, and cost-efficient environmental data.

CHAPTER 2

LITERATURE REVIEWS

Literature Review-1

Published By: International Journal of Application or Innovation in Engineering & Management (IJAIEEM).

Published On: July 2013

Authors : R.A.Roseline , Dr.M.Devapriya ,Dr.P.Sumathi

Pollution Monitoring Using Sensors and Wireless Sensor Networks

Present advances in electronic circuit miniature and micro-electromechanical systems (MEMS) have led to the creation of small sensor nodes which integrate several sensors, a central processing unit (CPU), memory and a wireless transceiver. Sensor networks [1,2] are a collection of these sensor nodes which are easily deployable and provide a high degree of visibility into real-world physical processes as they happen, thus benefitting a variety of applications as in Figure 1. Important applications include environmental and habitat monitoring, healthcare monitoring of patients, weather monitoring and forecasting. Military and homeland security surveillance, tracking of goods and manufacturing processes, safety monitoring of physical structures and construction sites, smart homes and offices, and many other uses that we do not yet imagine.



Figure 2.1 Wireless Sensor Network

ISSUES IN ENERGY CONSERVATION FOR SENSORS:

One of the biggest challenges in WSNs is the power issue, since sensors have limited source of power which is hard to replace or recharge. Typically, the batteries last only for a few months and the dead batteries pose an environmental hazard with the hardware left in the environment. Efforts to save power in sensors could be done using the following:

- 1) Reducing power consumption starts from the selection of Microcontroller unit (MCU), low power consumption MCU should be considered at first.
- 2) Choosing chip with low standby current and steady transreceiving current for Radio frequency module
- 3) Power source with low output voltage and low consumption power itself.
- 4) Reducing system operating frequency can lower consumption effectively.
- 5) Lowering system operating voltage influences system power consumption. So under the premise of system credibility, make sure that system is in lower operating voltage.

This survey has shown that the vast technological developments in wireless communication technology have led to the emergence of many Pollution monitoring sensors and wireless networks for monitoring and reporting pollution. This information could be used by the authorities to take necessary action such as emergency warning messages and evacuation of people to safe places. Further implementing pollution monitoring systems will help to assess how bad air pollution is from day to day and save the environment from further pollution episodes. This survey would help in the design and development of an energy efficient pollution monitoring system for India.

Literature Review-2

Published By: International Journal of Innovative Research in Science, Engineering and Technology

Published On: 4, April 2014

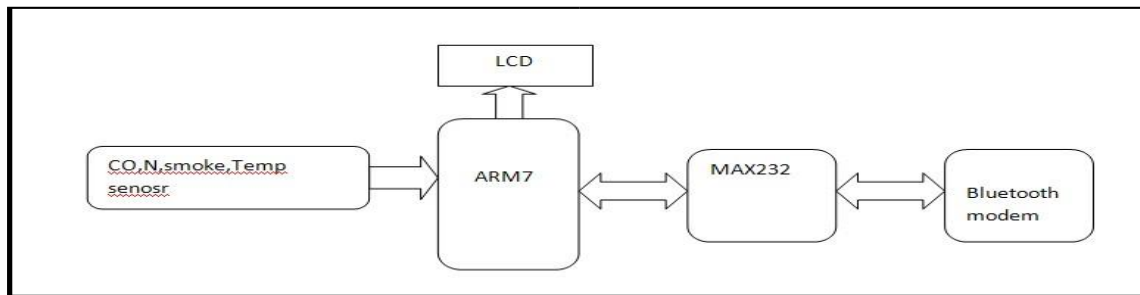
Authors : Gaikwad Varsha Bhagwan, Puranik V.G

Real Time Air Pollution Monitoring Using Mobile Phone

This system consists of air pollution sensors array, Global Positioning system (GPS) of mobile phone, single chip microcontroller, Bluetooth modem. Sensors are hardware devices that produce measurable response to a change in a physical condition of air pollution. The analog signal send by the sensors is digitized by an analog to digital converter & send to controller for further processing. CO, N, smoke and temp sensor senses the gas and communicate the data with microcontroller. Bluetooth modem is used for transmitting the data to LPC2138 to mobile. GPS modem of the mobile itself is used to send the location information. The main objective of this system is to continuously monitor the pollution level provided by the different sensor via Bluetooth modem at the control section. Hence particular plan of action should be taken to control the air pollution.



Figure 2.2: Smoke Releasing From Chimney

HARDWARE ARCHITECTURE:**Figure 2.3:** Central Control**A. Wireless sensor node**

A wireless sensor network (WSN) is an infrastructure consisting of computing, sensing and communication elements which allows the administrator to monitor & control of the specified parameters in the network. Typical application of WSN includes data collection, monitoring, surveillance & medical telemedicine. It is also used in irrigation system, Greenhouses for monitoring & controlling parameters like water flow, temp, humidity, moisture, etc.

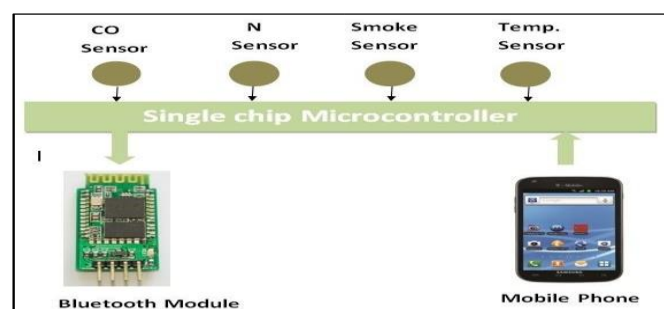
B. Sensors

Sensors are hardware devices that produce measurable response to a change in a physical condition of air pollution. The analog signal send by the sensors is digitized by an analog to digital converter & send to controller for further processing. There are different types of sensors from which we can select the suitable & appropriate sensor depending on the application.

Sensor 1 (CO sensor):- The CO sensor senses the gas and communicates the data with microcontroller.

Sensor 2 (N sensor):- The Nitrogen sensors sense the gas and communicate the data with microcontroller.

Sensor 3 (smoke sensor):- The smoke sensor senses the gas and communicates the data with microcontroller.

**Figure 2.4:** Mobile- DAC

This system consists of Mobile Data-Acquisition Unit (Mobile-DAQ) and fixed Internet Enabled Pollution monitoring Server(Pollution server).Mobile-DAQ consists of a 32-bit single-chip microcontroller integrated with a sensor array using analog ports. The Mobile-DAQ is connected to a Bluetooth module and mobile phone using RS-232 interface.

System utilizes city buses to collect pollutant gases such as CO, N, smoke and temp. The pollution data from various places sense by sensor arrays is transmitted to a control central unit that make this data available on the Internet through a Google Maps interface by using Bluetooth module.

SOFTWARE ARCHITECTURE

A mobile web server is software similar to Apache HTTP Server, or Apache Jetty Server, which provides a mobile device, such as tablet, smart phone. The capability to host web sites, web applications, and web services. Hosting web services on a mobile server will bring convenience, potential, flexibility to the consumers. This will revolutionize the way we share data, communicate and gather information. Mobile web servers face unlimited possibilities in the areas of enterprise mobile data access and sharing for lucrative markets such as the music, military and health care, are some of the examples. Engineering mobile devices around the world as mobile web servers would change the internet landscape. It would fundamentally alter the way we access web services such as maps with geo-location documents, medical records,information and news.Mass transit systems could use a mobile server hosting a geo-location service providing scheduling, real-time location and route information to the commuters.

Literature Review-3

Published By: International Journal of Engineering Trends and Technology.

Published On: 4- April 2013

Authors : P.Vijnatha Raju , R.V.R.S.Aravind , B Sangeeth Kumar

Pollution Monitoring System using Wireless Sensor Network in Visakhapatnam

As the technology increase, the degree of automation (minimizing the man power) in the almost all sectors are also increases. Wireless Sensor Networks (WSN) are gaining the ground in all sectors of life; from homes to factories, from traffic control to environmental monitoring. The air pollution monitoring system contains sensors to monitor the interested pollution parameter in environment. We simulated the three air pollutants gases including carbon monoxide, carbon dioxide & sulphur dioxide in air because these gases decides the degree of pollution level. We can also apply the approach in various applications like leaking cooking gas in our homes, to alert the workers in oil & gas industry to detect the leakage etc. This simulation creates the awareness in people in cities.

II. ZIGBEE STANDARD

The ZigBee standard is the new short range, low power, low data rate wireless networking technology for many real time application. It is best specified the bottom three layers (Physical, Data Link, and Network), as well an Application Programming Interface (API) based on the 7-layer OSI model for layered communication systems.

Figure shows the layered architecture adopted by the alliance industries. It should be noted that the ZigBee Alliance chose to use an already existing data link and physical layer specifications. These specifications are the published IEEE 802.15.4 standards for low rate personal area networks(PAN).



Figure 2.5: Zigbee layered Architecture

III PROBLEM ANALYSIS

Although the urbanization brings a higher economic development, the excessive population concentration will cause environmental damage and pollution like air pollution, noise pollution, water pollution etc. Due to leakage of gas in the real time applications like cooking (LPG) gas in our homes, leakage in oil & gas industries, leakage in pipelines of transfer of LPG gas & exposures to pollutants etc need to be detected & may results in the harm to human systems like nervous, cardiovascular systems.

Thus in our research , a real time monitoring of three gases are simulated in real environment. Gases that are monitored in our implementation are Carbon monoxide ,carbon dioxide & sulphur dioxide. In our simulation these three gases are successfully tested in four areas. We extended our simulated results to update in web.

IV. HARDWARE ARCHITECTURE

The proposed pollution node is designed by integrating the Sensor associate circuitry, Atmege 328p low power microcontroller and Xbee communication module. Design of Sensor node consist of 4 basic functionalities

a) Signal conditioning

The time gap between amount of gas concentration deposited on the sensing plates in heater and time requires to clear the gas concentration on the sensing plates.. Three sensor are used to detect the changes in gas concentration of various pollutions such as Carbon monoxide , carbon dioxide & sulphur concentration in air. As the output of the sensors are analog signals, strength and

correctness of signals need to assured.

b) Signal Amplification

As the signal detected by the sensors are need to be amplified and regenerated to increase the accuracy of the systems.

c) Signal calibration

ADC in Atmege 328p microcontroller provide the 10-bit resolution, provides the mapping between the analog input signal to digital signals for processing.

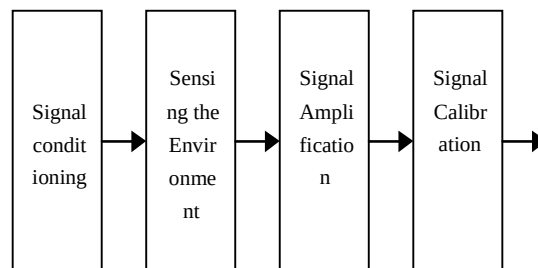


Figure 2.6: Design of pollution sensor node

Dependence & Power consumption of sensor nodes need to be minimized and functionality of each step should be optimized (maximize the cohesion and minimize the coupling). The selection of sensor and material used in construction of the sensor should select such that the there should be minimum changes in the accuracy of the system. As the increase of Carbon dioxide concentration(especially in urban area) and deforstization is a major reason to cause a global warning in addition to other pollutants. This paper describes implementation constraints and attributes or measure of the various pollution monitoring system using ZigBee technologies and embedded system using efficient way.

Literature Review-4

Published By: International Research Journal of Engineering and Technology (IRJET)

Published On: 09 , Sep 2020

Authors : Nikita D. Kadam, Rutuja N. Yadav, Aparna C. Pawar, A.G. Patil

IoT based Air Quality Monitoring and Controlling System using Raspberry Pi

This project describes the IOT Based Air Quality Monitoring and Controlling System Using Raspberry Pi. In previous paper the monitoring system is done with the help of raspberry pi. In some system GSM is used for information send but it not store large amount of data. The previous systems are only on the monitoring, so it not control the pollution. In this project we used raspberry pi, different sensors, ATmega328p-pu, firebese platform and air filter using these pollution is monitor and control. We can check pollution level on mobile/laptop through fire base platform where data is stored. Using different sensors connected to a Raspberry Pi module, environmental parameters like gas density, humidity, temperature and dust are observed and recorded at frequent intervals.

We can also control the quality of air pollution by using air filter, which absorb the carbon in the air and produce a fresh air.

A system which can monitor the leakage of toxic gases and hence the level of pollution using Raspberry-Pi and IoT is proposed which can prevent fatal accidents. By the use of MQ135/6/7 gas sensors the poisonous gases can be sensed and alert can be given to save the life of people. Raspberry-Pi serves as the heart of this module which controls the entire process. Wi-Fi module connects the whole process to internet and LCD is used for the visual Output. The use of wearable technology is also a mile stone which can ensure the safety of workers in the industrial floor. The air monitoring system can help in the innovation of new practices to overcome the problems of the highly-polluted areas, which is a major issue. It supports the new technology and effectively supports the healthy life concept.

CHAPTER 3

3.1 BLOCK DIAGRAM:

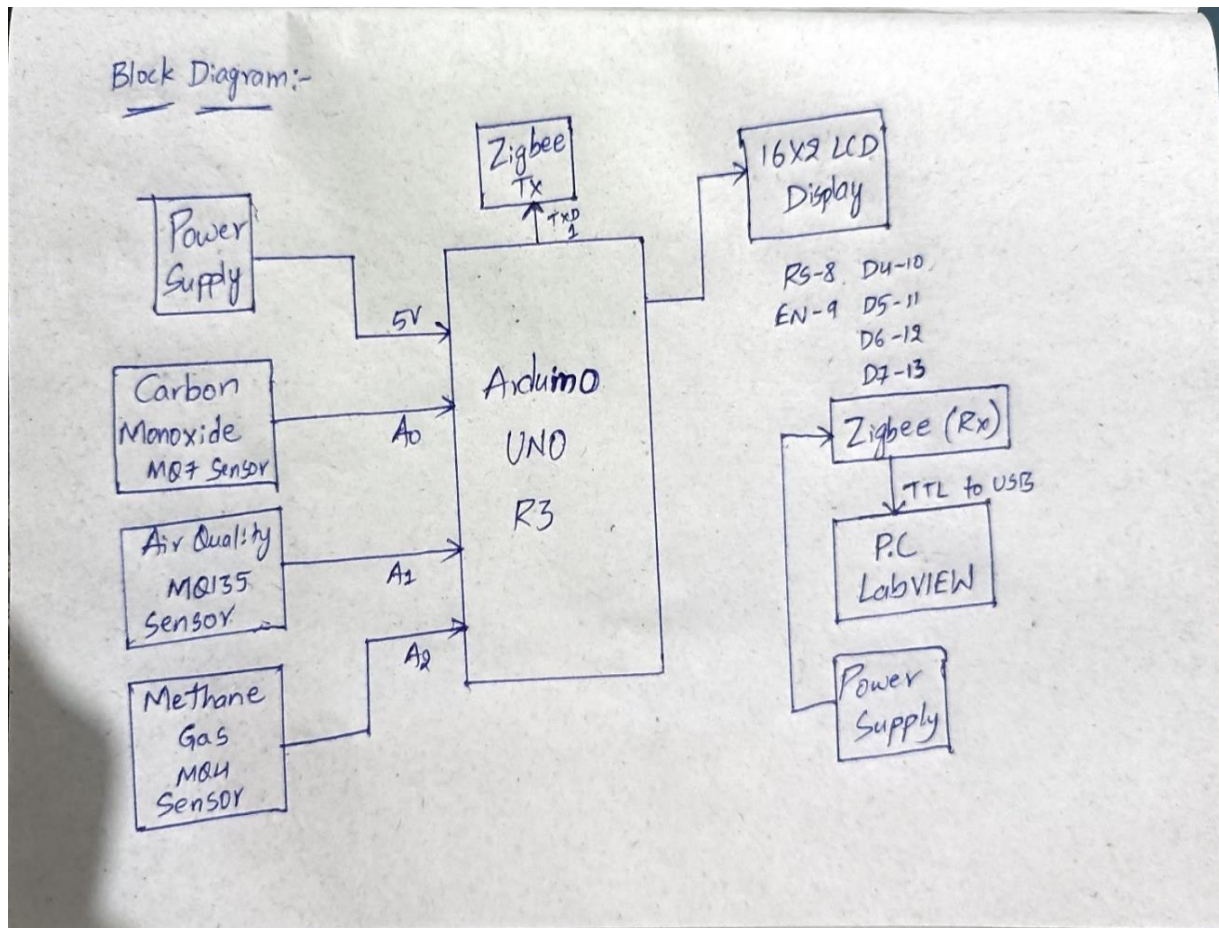


Figure 3.1: Block Diagram

This is a block diagram of a gas sensor system. The system consists of a Zigbee module, an LCD display, a power supply, an Arduino Uno microcontroller, a carbon monoxide sensor, a methane gas sensor, and a TTL to USB converter. The Zigbee module is used to communicate with the PC LabVIEW software. The LCD display is used to display the sensor readings. The power supply provides power to the system. The Arduino Uno microcontroller is used to control the system and process the sensor readings. The carbon monoxide sensor and the methane gas sensor are used to detect the presence of carbon monoxide and methane gas in the air. The TTL to USB converter is used to connect the Arduino Uno microcontroller to the PC LabVIEW software.

3.2 DATA FLOW DIAGRAM:

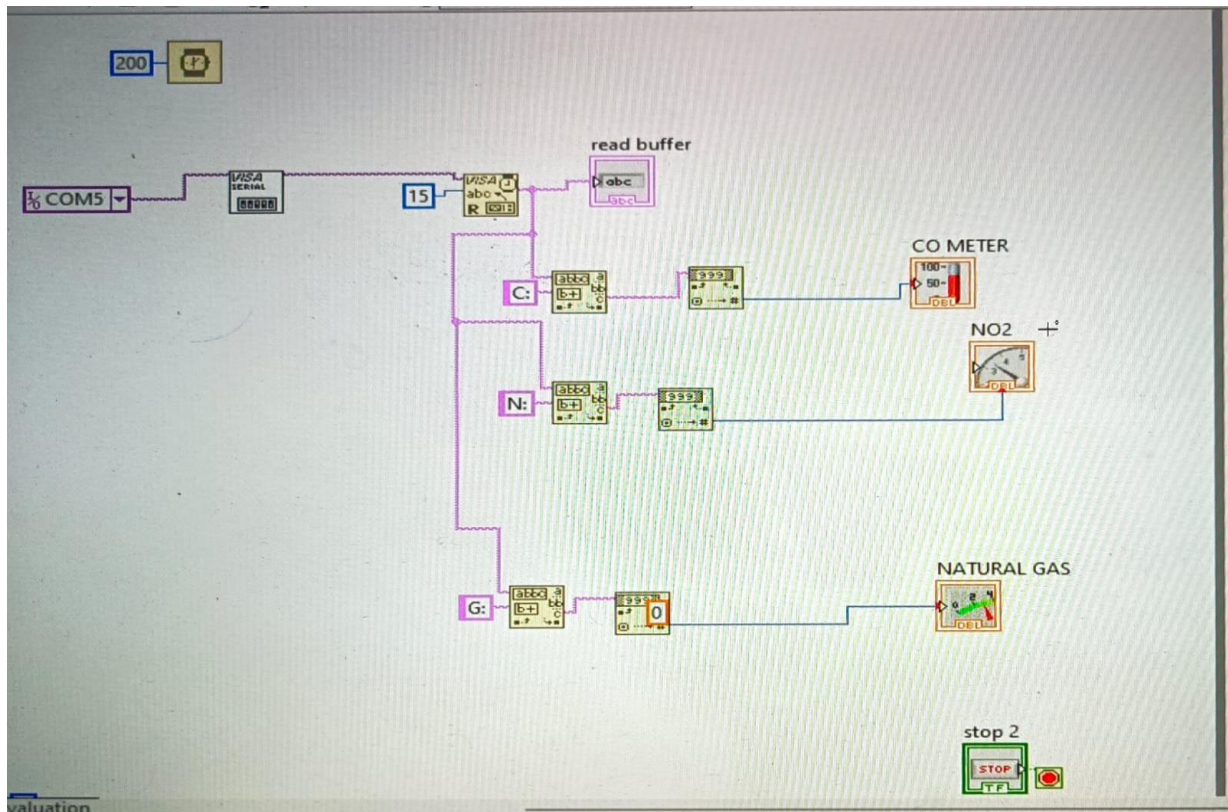


Figure 3.2: Data Flow Diagram in LabVIEW

3.3 REQUIRED COMPONENTS:

HARDWARE COMPONENTS:

1. ATMEGA16 AVR MICROCONTROLLER
2. ZIGBEE
3. MAX232
4. LCD DISPLAY
5. POWER SUPPLY
6. MQ7 GAS
7. MQ135
8. MQ4

SOFTWARE REQUIREMENTS:

1. AVRSTUDIO4 (IDE TOOL)
2. FLASH MAGIC PROGRAMMER
3. SERIAL COMMUNICATION PORT
4. SIMULATION TOOLS

3.4 APPLICATIONS:

1. HOME AUTOMATIONS
2. INDUSTRIAL CONTROLS
3. CONSUMER ELECTRONICS.

3.5 PROPOSED SYSTEM

The proposed system for industrial pollution monitoring using LabVIEW and Arduino integrates multiple gas sensors, including the MQ7 for carbon monoxide, MQ135 for air quality, and MQ4 for methane gas detection. The sensors are connected to an Arduino Uno microcontroller, which collects and processes data in real-time.

Through LabVIEW software on a PC, users can monitor pollution levels remotely via Zigbee communication. The system features a user-friendly 16x2 LCD display for on-site data visualization and analysis.

This comprehensive solution enables proactive detection and analysis of industrial pollutants, facilitating timely interventions to mitigate environmental risks and ensure community well-being.

CHAPTER 4

4.1 INTRODUCTION TO EMBEDDED SYSTEMS:

Embedded systems are electronic devices that incorporate microprocessors with in Their implementations. The main purposes of the microprocessors are to simplify the system design and provide flexibility. Having a microprocessor in the device means that removing the bugs, making modifications, or adding new features are only matters of rewriting the software that controls the device. Or in other words embedded computer systems are electronic systems that include a microcomputer to perform a specific dedicated application. The computer is hidden inside these products. Embedded systems are ubiquitous. Every week millions of tiny computer chips come pouring out of factories finding their way into our everyday products.

Embedded systems are self-contained programs that are embedded within a piece of hardware. Whereas a regular computer has many different applications and software that can be applied to various tasks, embedded systems are usually set to a specific task that cannot be altered without physically manipulating the circuitry. Another way to think of an embedded system is as a computer system that is created with optimal efficiency, thereby allowing it to complete specific functions as quickly as possible.

Embedded systems designers usually have a significant grasp of hardware technologies. They used specific programming languages and software to develop embedded systems and manipulate the equipment. When searching online, companies offer embedded systems development kits and other embedded systems tools for use by engineers and businesses.

Embedded systems technologies are usually fairly expensive due to the necessary development time and built in efficiencies, but they are also highly valued in specific industries. Smaller businesses may wish to hire a consultant to determine what sort of embedded systems will add value to your organization.

Characteristics:

Two major areas of differences are cost and power consumption. Since many embedded systems are produced in the tens of thousands to millions of units range, reducing cost is a major concern. Embedded systems often use a (relatively) slow processor and small memory size to minimize costs.

The slowness is not just clock speed. The whole architecture of the computer is often intentionally simplified to lower costs. For example, embedded systems often use peripherals controlled by synchronous serial interfaces, which are ten to hundreds of times slower than comparable peripherals used in PCs.

Programs on an embedded system often must run with real-time constraints with limited hardware resources: often there is no disk drive, operating system, keyboard or screen.

A flash drive may replace rotating media, and a small keypad and LCD screen may be used instead of a PC's keyboard and screen.

Firmware is the name for software that is embedded in hardware devices, e.g. in one or more ROM/Flash memory IC chips.

Embedded systems are routinely expected to maintain 100% reliability while running continuously for long periods, sometimes measured in years. Firmware is usually developed and tested to much stricter requirements than is general-purpose software, which can usually be easily restarted if a problem occurs.

4.2 EMBEDDED SYSTEMS:

An **embedded system** is a special-purpose system in which the computer is completely encapsulated by or dedicated to the device or system it controls. Unlike a general-purpose computer, such as a personal computer, an embedded system performs one or a few predefined tasks, usually with very specific requirements. Since the system is dedicated to specific tasks, design engineers can optimize it, reducing the size and cost of the product. Embedded systems are often mass-produced, benefiting from economies of scale. Personal digital assistants (PDAs) or handheld computers are generally considered embedded devices because of the nature of their hardware design, even though they are more expandable in software terms. This line of definition continues to blur as devices expand. Physically, embedded systems range from portable devices such as digital watches and MP3 players, to large stationary installations like traffic lights, factory controllers, or the systems controlling nuclear power plants. In terms of complexity embedded systems can range from very simple with a single microcontroller chip, to very complex with multiple units, peripherals and networks mounted inside a large chassis or enclosure.

4.3 EXAMPLES OF EMBEDDED SYSTEMS:

- Automatic teller machines (ATMs)
- Avionics, such as inertial guidance systems, flight control hardware/software and other integrated systems in aircraft and missiles
- Cellular telephones and telephone switches
- Engine controllers and antilock brake controllers for automobiles
- Home automation products, such as thermostats, air conditioners, sprinklers, and security monitoring systems
- Handheld calculators, Handheld computers
- Household appliances, including microwave ovens, washing machines, television sets, DVD players and recorders
- Medical equipment
- Personal digital assistant, Videogame consoles
- Computer peripherals such as routers and printers
- Industrial controllers for remote machine operation.

4.4. INTRODUCTION TO ZIGBEE:

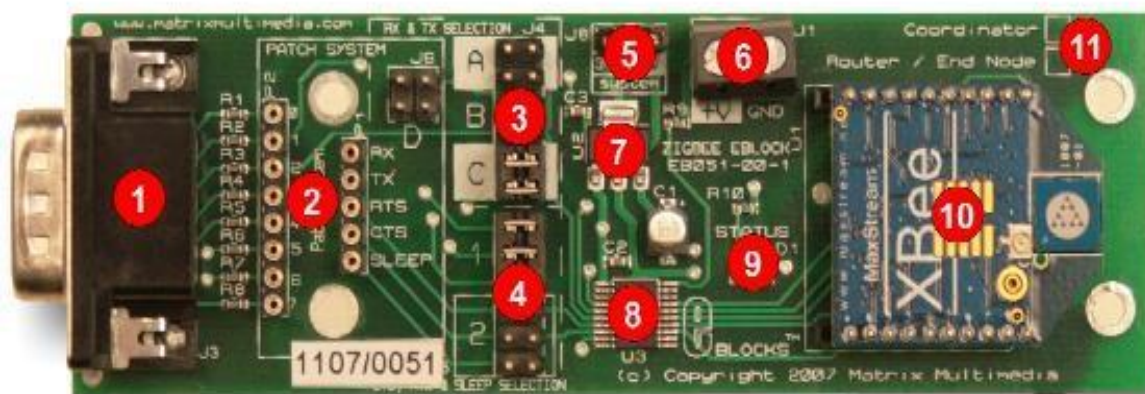


Figure 4.1: Zigbee Chipset

4.5 DESCRIPTION

Provides a Zigbee node interface that can connect to or create a Zigbee network. EB051C – Coordinator Zigbee node, used to start, configure the network and allow other nodes to join. EB051R – Router / End device node, used to connect and communicate to networks started by a EB051C. Zigbee is a software-based protocol that sits on top of the 802.11 RF wireless

devices standard similar to Bluetooth. Unlike Bluetooth, Zigbee is capable of forming large networks of nodes and boasts advanced features such as mesh networking, simple addressing structures, route detection, route repair, guaranteed delivery and low power operation Modes.

The EB051 Zigbee E-Blocks are fully compliant with both the Zigbee pro and ZNET

Zigbee standards. The boards can be used create a network of dynamic moveable Zigbee nodes, or to interface with an existing Zigbee network. Zigbee provides a transparent layer for sending and receiving data from the network. Therefore once the module has been configured and assigned to the correct address then sending and receiving data is as simple as sending and Receiving RS232 bytes through the chip's UART.

4.6 FEATURES OF ZIGBEE

- Zigbee Wireless communications
- Flow code macros available
- Compatible with global RF standards
- Onboard Zigbee module
- Status LED
- Full 2007 Zigbee Pro / ZNET compliance
- 128-bit AES Encryption
- Range of approx 100m per node

4.7 ZIGBEE OPERATION :

The Zigbee boards use a V2 XBEE module to interface to the Zigbee network. These modules are compliant with the 2007 Zigbee Pro / ZNET standard. The V2 XBEE modules come in two varieties. One is configured to be the Zigbee network coordinator (EB051C) and the other is configured to be either a router node or an end device node (EB051R). The variety of the module is marked at the top right hand side of the Zigbee board. Coordinator nodes are responsible for creating the Zigbee network and allowing other Zigbee nodes to join. Only one coordinator node can exist on any single network. Router nodes are responsible for routing signals to other routers or to end nodes. End device nodes are responsible for collecting or depositing real world data to and from the Zigbee network. The Coordinator node and Router nodes are capable of handling up to eight children devices. The children devices can consist of

either other Router nodes or End device nodes. If an End device node is configured to sleep then the parent device associated for that node will be responsible for buffering any incoming data. Therefore if you are using sleeping End devices you must make sure to poll the parent for data every time the device comes out of sleep mode.

3.3V system compatibility The board is compatible with 3.3V and 5V systems.

4.8 COMMUNICATIONS :

The XBEE modules are configured by means of using a TTL level RS232 bus to send and receive AT commands. This protocol requires a start bit, eight data bits and a stop bit. The baud rate for the XBEE modules is set to 9600, with no parity and flow control lines RTS and CTS that can be used. AT commands are strings of ASCII data that are sent over the RS232 bus. For more information on the AT commands used by the XBEE module please refer to the V2 XBEE datasheet. Example AT command ATID 234 - Assigns a personal area network identifier of 0x234 or 564 in decimal.

6. V2 XBEE Module For further data regarding the XBEE module please visit the following link.

4.9 BOARD LAYOUT :

- 9 Way D-type Plug
- Patch system
- Rx and Tx UART routing selection
- CTS, RTS and SLEEP routing selection
- Input supply voltage select
- Input supply voltage screw terminals
- 3.3V regulator
- Voltage level shifter
- Status LED
- XBEE Zigbee module
- Coordinator / Router / End device marking.

ZigBee is a wireless network protocol specifically designed for low rate sensor and control networks. Compared to other wireless protocols, the ZigBee wireless protocol offers

- low complexity,

- reduced resource requirements
- And most importantly, a standard set of specifications.
- It also offers three frequency bands of operation along with a number of network configurations
- And optional security capability.

4.10 ZIGBEE

Performance:

XBee

- Power output:
- 2mW (+3 dBm) boost mode
- 1.25 mW (+1 dBm) normal mode
- Indoor/Urban range: Up to 133 ft (40 m)
- Outdoor/RF line-of-sight range: Up to 400 ft (120 m)
- Operating frequency: 2.4 GHz
- Receiver sensitivity:
- 96 dBm boost mode
- 95 dBm normal mode

Performance: XBee-PRO

- Power output:
- 50 mW (+17 dBm) North American version
- 10 mW (+10 dBm) International version
- Indoor/Urban range: Up to 400 ft (120 m)
- Outdoor/RF line-of-sight range: Up to 1 mile (1.6 km) RF LOS
- RF data rate: 250 Kbps
- Interface data rate: Up to 1 Mbps software selectable
- Operating frequency: 2.4 GHz
- Receiver sensitivity: -102 dBm (all variants)

Networking:

- Spread Spectrum type: DSSS (Direct Sequence Spread Spectrum)
- Networking topology: Mesh, point-to-point & point-to-multipoint

- Error handling: Retries & acknowledgements
- Filtration options: PAN ID, Channel, and 64-bit addresses
- Channel capacity:
- XBee: 16 Channels
- XBee-PRO: 13 Channels
- Addressing: 65,000 network addresses available for each channel

Power:

Supply voltage:

- XBee: 2.1 - 3.6 VDC
- XBee-PRO: 3.0 - 3.4 VDC
- XBee Footprint Recommendation: 3.0 - 3.4 VDC

4.11 TARANG MODULE

Features and Benefits:

- Point to multi point and peer-to-peer topologies on proprietary stack.
- Direct Sequence Spread Spectrum technology.
- Each direct sequence channel has 64K unique network addresses
- Transmit Power: 0 dBm.
- RF data rate: 250 kbps.
- Acknowledgement mode communication with retries.
- Power saving modes.
- Source / destination addressing.
- Unicast and broadcast communication.
- Digital I/O line support.
- Default configuration for ready to use.

4.11.1. TARANG

Tarang modules are designed with low to medium transmit power and for high reliability wireless networks. The modules require minimal power and provide reliable delivery of data between devices. The interfaces provided with the module help to directly fit into many industrial applications. The modules operate within the ISM 2.4-2.4835 GHz frequency band with IEEE 802.15.4 baseband.

4.11.2 FEATURES

- Range - Outdoor line of sight: up to 50kms with directional antenna.
- Transmit Power: up to 1 watt / 30 dBm nominal.
- Receiver Sensitivity: up to -107 dBm.
- AT Command Modes for configuring Module Parameters
- Direct sequence spread spectrum technology.
- Analog to digital conversion and digital I/O line support

4.11.3 TARANGNET FEATURES

- Acknowledgement mode communication with retries
- Each direct sequence channel has 64K unique network addresses.
- Unicast and broadcast communication.

4.12 ZIGBEE PIN CONFIGURATION

Pin No.	Name	Direction	Description
1	VCC	Input	Power Supply
2	DOUT	Output	Serial Data Out
3	DIN	Input	Serial Data In
4	RESERVED	-	
5	RST	Input	Module Reset
6	*RSSI	Output	RSSI Indicator
7	*PWM	Output	PWM Output
8	BGND	Input	Programming Pin
9	SLEEP	Input	Sleep Control
10	GND	-	Ground
11	AD4/DIO4	I/O	Analog Input 4 or Digital I/O 4
12	CTS/DIO7	I/O	CTS or Digital I/O 7
13	*STATUS	Output	Module Status
14	VREF	Input	Reference Voltage for Analog Input
15	AD5/DIO5	I/O	Analog Input 5 or Digital I/O 5
16	RTS/DIO6	I/O	RTS or Digital I/O 6
17	AD3/DIO3	I/O	Analog Input 3 or Digital I/O 3
18	AD2/DIO2	I/O	Analog Input 2 or Digital I/O 2
19	AD1/DIO1	I/O	Analog Input 1 or Digital I/O 1
20	AD0/DIO0	I/O	Analog Input 0 or Digital I/O 0

Figure 4.2: Zigbee Pin Configuration

4.12.1 INTERFACE AND OPERATION

The Tarang modules interface to a host device through a logic-level asynchronous serial port. Through its serial port, the module can communicate with any logic and voltage compatible UART or through a level translator to any serial device (For example: RS-232 or USB interface board).

4.12.2 SERIAL INTERFACE

Tarang can be interfaced with a micro controller or a PC using serial port with the help

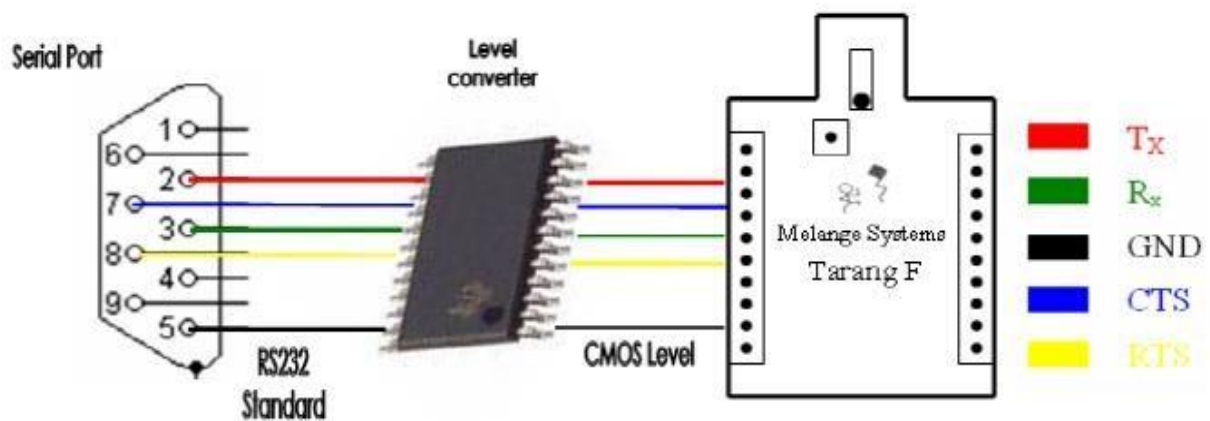


Figure 4.3: Serial Interface

of CTS and RTS are optional. (Refer pin configuration for pin details) Tarang supports serial data with,

- Flow Control: Hardware, None
- Parity: None
- Baud Rates: 1200,2400,4800,9600,19200,38400,57600,115200
- Data Bits: 8

To establish a successful serial communication with the module, serial parameters need to be configured properly in the module and host side. Both the module and PC settings can be Viewed and set using AT command set through popular terminal applications like 'HyperTerminal'.

CHAPTER 5

5.1 ANALOG TO DIGITAL CONVERTER:

- 10-bit Resolution
- 0.5 LSB Integral Non-linearity
- ± 2 LSB Absolute Accuracy
- 13 - 260 μ s Conversion Time
- Up to 15 kSPS at Maximum Resolution
- 8 Multiplexed Single Ended Input Channels
- 7 Differential Input Channels
- 2 Differential Input Channels with Optional Gain of 10x and 200x
- Optional Left adjustment for ADC Result Readout
- 0 - VCC ADC Input Voltage Range
- Selectable 2.56V ADC Reference Voltage
- Free Running or Single Conversion Mode
- ADC Start Conversion by Auto Triggering on Interrupt Sources
- Interrupt on ADC Conversion Complete

The ATmega16 features a 10-bit successive approximation ADC. The ADC is connected to an 8-channel Analog Multiplexer which allows 8 single-ended voltage inputs constructed from the Pins of Port A. The single-ended voltage inputs refer to 0V (GND).

The device also supports 16 differential voltage input combinations. Two of the differential inputs (ADC1, ADC0 and ADC3, ADC2) are equipped with a programmable gain stage, providing amplification steps of 0 dB (1x), 20 dB (10x), or 46 dB (200x) on the differential input voltage before the A/D conversion. Seven differential analog input channels share a common negative terminal (ADC1), while any other ADC input can be selected as the positive input terminal. If 1x or 10x gain is used, 8-bit resolution can be expected. If 200 x gains is used, 7-bit resolution can be expected.

The ADC contains a Sample and Hold circuit which ensures that the input voltage to the ADC is Held at a constant level during conversion.

The ADC has a separate analog supply voltage pin, AVCC. AVCC must not differ more than ± 0.3 V from VCC. See the paragraph “ADC Noise Canceler” on how to connect this Pin.

5.2 ANALOG TO DIGITAL CONVERTER BLOCK SCHEMATIC

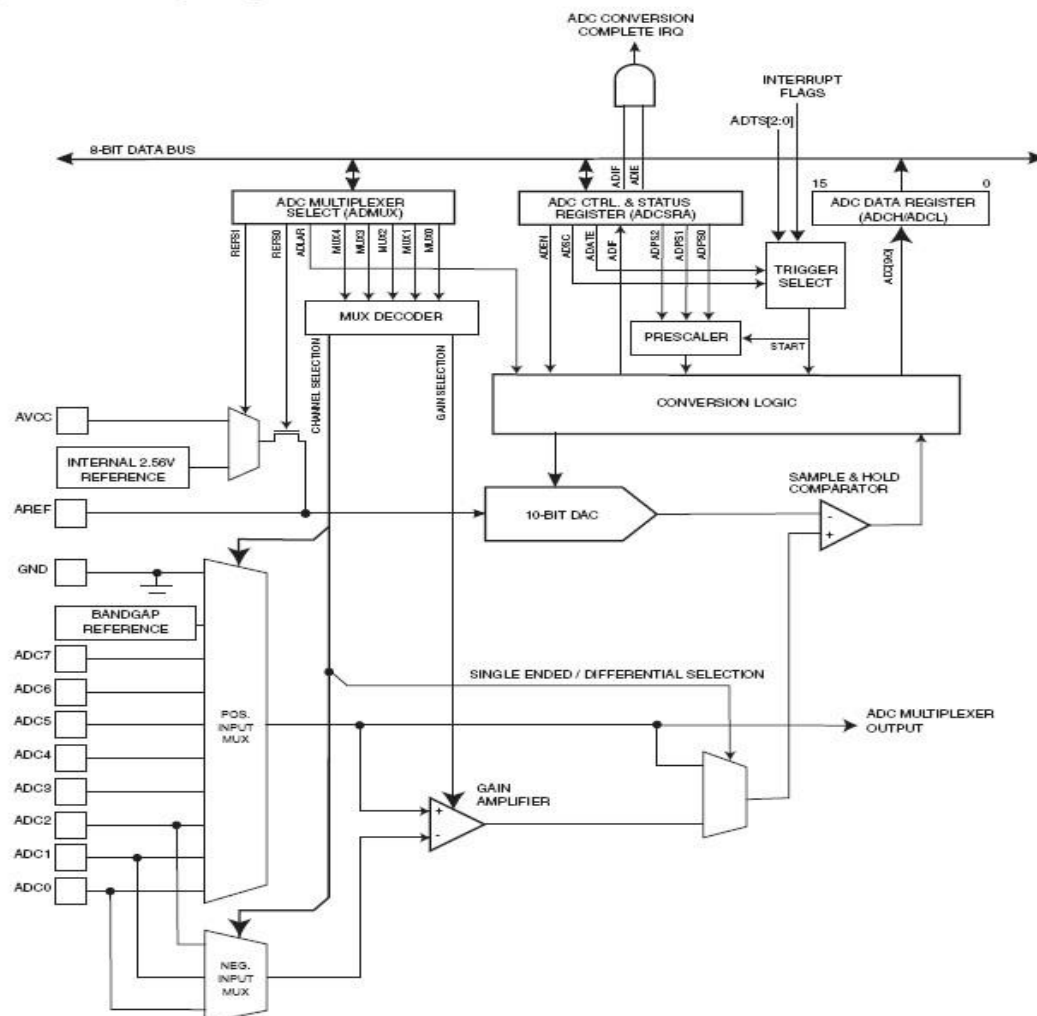


Fig 5.1: Analog to Digital Converter Block Schematic

5.3 OPERATION:

The ADC converts an analog input voltage to a 10-bit digital value through successive approximation. The minimum value represents GND and the maximum value represents the voltage on the AREF pin minus 1 LSB. Optionally, AVCC or an internal 2.56V reference voltage may be connected to the AREF pin by writing to the REFSn bits in the ADMUX Register. The internal voltage reference may thus be decoupled by an external capacitor at the AREF pin to improve noise immunity.

The analog input channel and differential gain are selected by writing to the MUX bits in

ADMUX. Any of the ADC input pins, as well as GND and a fixed bandgap voltage reference, can be selected as single ended inputs to the ADC. A selection of ADC input pins can be selected as positive and negative inputs to the differential gain amplifier. If differential channels are selected, the differential gain stage amplifies the voltage difference between the selected input channel pair by the selected gain factor. This amplified value then becomes the analog input to the ADC. If single ended channels are used, the gain amplifier is bypassed altogether.

The ADC is enabled by setting the ADC Enable bit, ADEN in ADCSRA. Voltage reference and Input channel selections will not go into effect until ADEN is set. The ADC does not consume Power when ADEN is cleared, so it is recommended to switch off the ADC before entering power saving sleep modes.

The ADC generates a 10-bit result which is presented in the ADC Data Registers, ADCH and ADCL. By default, the result is presented right adjusted, but can optionally be presented left Adjusted by setting the ADLAR bit in ADMUX. If the result is left adjusted and no more than 8bit precision is required, it is sufficient to read ADCH. Otherwise, ADCL must be read first, then ADCH, to ensure that the content of the Data Registers belongs to the same conversion. Once ADCL is read, ADC access to Data Registers is blocked. This means that if ADCL has been read, and a conversion completes before ADCH is read, neither register is updated and the result from the conversion is lost. When ADCH is read, ADC access to the ADCH and ADCL Registers is reenabled. The ADC has its own interrupt which can be triggered when a conversion completes. When ADC access to the Data Registers is prohibited between reading of ADCH and ADCL, the interrupt will trigger even if the result is lost.

5.4 STARTING A CONVERSION:

A single conversion is started by writing a logical one to the ADC Start Conversion bit, ADSC. This bit stays high as long as the conversion is in progress and will be cleared by hardware when the conversion is completed. If a different data channel is selected while a conversion is in progress, the ADC will finish the current conversion before performing the channel change. Alternatively, a conversion can be triggered automatically by various sources. Auto Triggering is enabled by setting the ADC Auto Trigger Enable bit, ADATE in ADCSRA. The trigger source is selected by setting the ADC Trigger Select bits; ADTS in SFIOR (see description of the ADTS bits for a list of the trigger sources). When a positive edge occurs on the selected trigger signal, the ADC prescaler is reset and a conversion is started. This provides a method of starting conversions at fixed intervals. If the trigger signal still is set when the conversion completes, a

new conversion will not be started. If another positive edge occurs on the trigger signal during conversion, the edge will be ignored. Note that an Interrupt Flag will be set even if the specific interrupt is disabled or the global interrupt enable bit in SREG is cleared. A conversion can thus be triggered without causing an interrupt. However, the Interrupt Flag must be cleared in order to trigger a new conversion at the next interrupt event

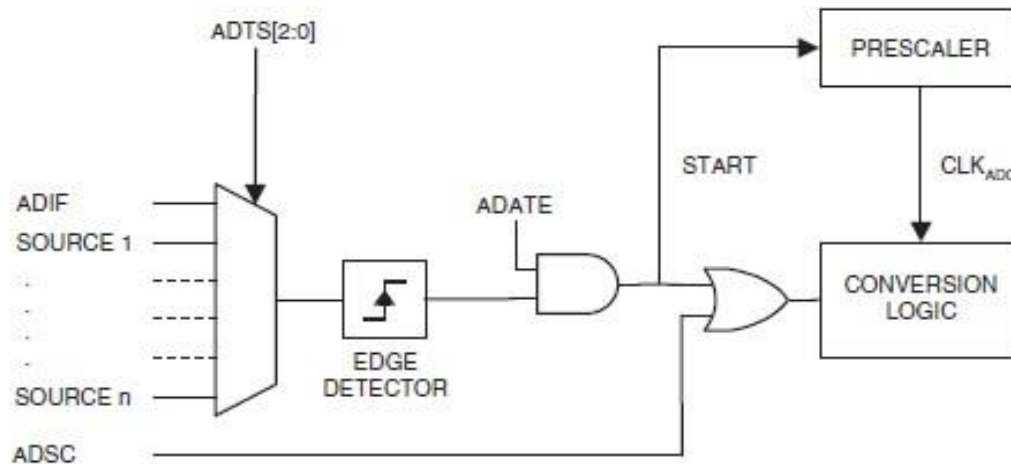


Figure 5.2: Process of Conversion

Using the ADC Interrupt Flag as a trigger source makes the ADC start a new conversion as soon as the ongoing conversion has finished. The ADC then operates in Free Running mode, constantly sampling and updating the ADC Data Register. The first conversion must be started by writing a logical one to the ADSC bit in ADCSRA. In this mode the ADC will perform successive conversions independently of whether the ADC Interrupt Flag, ADIF is cleared or not. used to determine if a conversion is in progress.

CHAPTER 6

HARDWARE CONSTRAINTS

6.1 LIQUID CRYSTAL DISPLAY:

Liquid crystal display a type of display used in digital watches and many portable computers. LCD displays utilize two sheets of polarizing material with a liquid crystal solution between them. An electric current passed through the liquid causes the crystals to align so that light cannot pass through them. Each crystal, therefore, is like a shutter, either allowing light to pass through or blocking the light. LCDs have become very popular over recent years for information display in many smart 'appliances. They are usually controlled by microcontrollers. They make complicated equipment easier to operate. LCDs come in many shapes and sizes but the most common is the 20 character x 4 line display with no backlight. It requires only 11 connections – eight bits for data (which can be reduced to four if necessary) and three control lines (we have only used two here). It runs off a 5V DC supply and only needs about 1mA of current. The display contrast can be varied by changing the voltage into pin 3 of the display, usually with a trim pot.

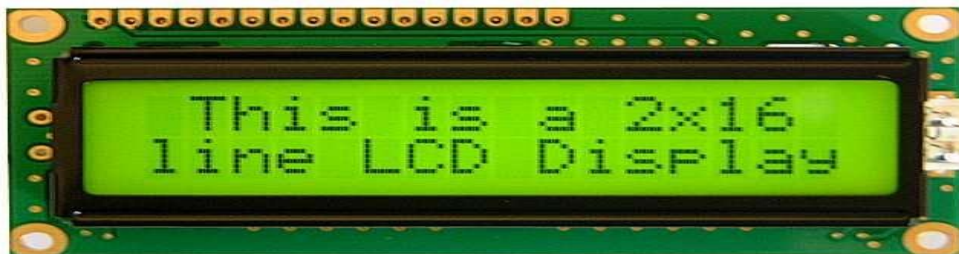


Figure 6.1: LCD Display

LCD adds a lot to our application in terms of providing a useful interface for the user, debugging an application or just giving it a "Professional" look. The most common type of LCD controller is the Hitachi 44780, which provides a relatively simple interface between a processor and an LCD.

6.2 DESCRIPTION FOR INTERFACING OF LCD:

To get the display working requires eight bits of data, a register select line (RS) and a strobe line (E). A third input, R/W, is used to read or write data to/from the LCD. The eight bits of data are supplied from the controller port data lines and two controller port control lines are used for RS ('auto') and E ('strobe'). Basically the LCD has two registers, a data register and a command register. Data is written into the command register when RS is low and into the data register when RS is high. Data is latched into the LCD register on the falling edge of Enable'.

6.3 PIN DESCRIPTION OF LCD:

Pins	Description
1	Ground
2	Vcc
3	Contrast Voltage
4	"R/S" _Instruction/Register Select
5	"R/W" _Read/Write LCD Registers
6	"E" Clock
7 - 14	Data I/O Pins

Figure 6.2: Pin Description of LCD

From this description, the interface is a parallel bus, allowing simple and fast reading/writing of data to and from the LCD. This waveform will write an ASCII Byte out to the LCD's screen. The ASCII code to be displayed is eight bits long and is sent to the LCD either four or eight bits at a time. If four bit mode is used, two "nibbles" of data (Sent high four bits and then low four bits with an "E" Clock pulse with each nibble) are sent to make up a full eight bit transfer. The "E" Clock is used to initiate the data transfer within the LCD.

Sending parallel data, as either four or eight bits are the two primary modes of operation. While there are secondary considerations and modes, deciding how to send the data to the LCD is most critical decision to be made for an LCD interface application. Eight-bit mode is best used when speed is required in an application and at least ten I/O pins are available. Four bit mode requires a minimum of six bits. To wire a microcontroller to an LCD in four bit mode, just the top four bits (DB4-7) are written to. The "R/S" bit is used to select whether data or an instruction is being transferred between the microcontroller and the LCD. If the Bit is set, then the byte at

the current LCD "Cursor" Position can be reader written. When the Bit is reset, either an instruction is being sent to the LCD or the execution status of the last instruction is read back (whether or not it has completed).

6.4 SENSORS:

6.4.1 CARBON MONOXIDE SENSOR - MQ-7:



Figure 6.3: Carbon Monoxide Sensor - MQ-7

Description: This is a simple-to-use [Carbon Monoxide \(CO\)](#) sensor, suitable for sensing CO concentrations in the air. The MQ-7 can detect CO-gas concentrations anywhere from 20 to 2000ppm. This sensor has a high sensitivity and fast response time. The sensor's output is an analog resistance. The drive circuit is very simple; all you need to do is power the heater coil with 5V, add a load resistance, and connect the output to an ADC. This sensor comes in a package similar to our MQ-3 alcohol sensor, and can be used with the breakout board below.

TECHNICAL DATA MQ-7 GAS SENSOR :

FEATURES

- * High sensitivity to carbon monoxide
- * Stable and long life

APPLICATION

They are used in gas detecting equipment for carbon monoxide(CO) in family and industry or car.

SPECIFICATIONS

- Power supply needs: 5V
- Interface type: Analog
- Pin Definition: 1-Output 2-GND 3-VCC

- High sensitivity to carbon monoxide
- Fast response
- Stable and long life
- Size: 40x20mm

6.4.2 AIR QUALITY SENSOR- MQ135:



Figure 6.4: Air quality Sensor- MQ135

The MQ-135 sensor is a type of gas sensor that is commonly used for detecting a variety of gases in the air. It is particularly known for its ability to sense and measure the concentration of gases such as carbon dioxide (CO₂), ammonia (NH₃), methane (CH₄), benzene, and various other volatile organic compounds (VOCs).

Here are some key features and characteristics of the MQ-135 sensor:

Detection Range: The MQ-135 sensor is sensitive to a range of gases, making it versatile for different applications. It is often used for monitoring air quality and detecting the presence of harmful gases.

Principle of Operation: The sensor operates on the principle of resistance change. It contains a tin dioxide (SnO₂) sensing layer that reacts with gases in the air, leading to a change in the sensor's resistance. The extent of this resistance change is proportional to the concentration of the gas.

Analog Output: The MQ-135 sensor provides analog output signals that can be read by a microcontroller or other data acquisition systems. The voltage across the sensor's analog output pin can be used to determine the concentration of the detected gas.

SPECIFICATIONS:

Sensor Type: Semiconductor gas sensor.

Target Gases: Commonly sensitive to gases such as carbon dioxide (CO₂), ammonia (NH₃), methane (CH₄), benzene, and various volatile organic compounds (VOCs).

Sensitivity Adjustment: The sensor may have a sensitivity adjustment feature, allowing users to calibrate the sensor for specific applications.

Operating Voltage: Typically operates at 5V DC.

Heating Element Voltage: Requires a heater voltage for the preheating of the sensor element. The recommended voltage is often around 5V.

Power Consumption: The power consumption is relatively low, making it suitable

6.4.3 METHANE GAS SENSOR – MQ4



Figure 6.5: Methane Gas Sensor- MQ4

Type: The MQ-135 is a semiconductor gas sensor that operates on the principle of resistance changes in response to the presence and concentration of certain gases.

Target Gases: It is designed to detect a range of gases, including carbon dioxide (CO₂), ammonia (NH₃), methane (CH₄), benzene, and various volatile organic compounds (VOCs).

Sensing Mechanism: The sensor contains a tin dioxide (SnO₂) sensing layer. When this layer comes into contact with the target gases, it undergoes a change in resistance, which is then measured to determine gas concentration.

Operating Voltage: Typically operates at 5V DC. It may have a separate heater element that requires an additional voltage (usually around 5V) for preheating.

Heating Element: The sensor has a built-in heating element that needs to be powered for a certain period before obtaining stable and accurate readings. This preheating is crucial for the

sensor's proper functioning.

Output Signal: Provides an analog output voltage that varies depending on the concentration of the detected gases. This analog signal can be interfaced with a microcontroller or other electronics for further processing.

Sensitivity Adjustment: Some versions of the MQ-135 sensor may include a sensitivity adjustment feature, allowing users to fine-tune the sensor's response based on specific application requirements.

Applications: Commonly used in air quality monitoring devices, gas leak detectors, and environmental monitoring systems. It is suitable for indoor air quality assessment and can be employed in various DIY projects.

6.5 ARDUINO UNO:



Figure 6.6: Arduino UNO

The Arduino Uno is a popular and versatile microcontroller board designed for easy prototyping and development of electronic projects. Here's a brief description of its key features:

Microcontroller: The Arduino Uno is powered by the ATmega328P microcontroller, which provides ample computing power for a wide range of applications.

Digital and Analog Pins: It features 14 digital input/output pins that can be used for connecting sensors, actuators, and other electronic components.

Additionally, there are 6 analog input pins for reading analog sensor values.

Programmability: The board can be programmed using the Arduino Integrated Development Environment (IDE), simplifying the coding process for both beginners and experienced developers.

Clock Speed: The ATmega328P operates at a clock speed of 16 MHz, enabling relatively fast execution of instructions.

USB Connectivity: Arduino Uno has a built-in USB interface, allowing easy communication with a computer for programming and serial communication.

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Clock Speed: The ATmega328P operates at a clock speed of 16 MHz, enabling relatively fast execution of instructions.

USB Connectivity: Arduino Uno has a built-in USB interface, allowing easy communication with a computer for programming and serial communication.

Power Options: It can be powered through the USB connection or an external power source, providing flexibility in various project scenarios.

Voltage Regulator: Equipped with a voltage regulator, the Arduino Uno can accept a wide range of input voltages, making it compatible with various power sources.

Flash Memory: The ATmega328P has 32 KB of flash memory, where users can store their program code.

EEPROM and SRAM: It includes 1 KB of EEPROM for non-volatile data storage and 2 KB of SRAM for temporary data storage during program execution.

Open-Source Platform: Arduino Uno is part of the open-source Arduino platform, encouraging collaboration and the sharing of designs and code among the global community.

Community Support: With a large and active community, users can find extensive resources, tutorials, and forums for troubleshooting and expanding their knowledge.

Easy to Use: Arduino is designed with beginners in mind. It has a simple programming environment and a user-friendly Integrated Development Environment (IDE) that allows users to write, compile, and upload code to the Arduino board easily.

Variety of Boards: Arduino offers a variety of boards catering to different needs. Common boards include the Arduino Uno, Arduino Nano, Arduino Mega, and more, each with its own set of features and capabilities.

Expansion Headers/Pins: Arduino boards typically have a set of digital and analog pins that can be used to connect various sensors, actuators, and other components. These pins allow for easy interfacing with external devices.

Shield Compatibility: Arduino boards support the use of "shields," which are add-on boards that can be stacked on top of the Arduino to provide additional functionality. There are shields for various purposes, such as Ethernet connectivity, Wi-Fi, motor control, and more.

Wide Range of Libraries: Arduino has a vast library of pre-written code that makes it easier to interface with sensors, displays, communication modules, and other components. This simplifies the coding process and accelerates project development.

Cross-Platform Compatibility: The Arduino IDE is available for multiple operating systems, including Windows, macOS, and Linux, making it accessible to a broad user base.

Operation of Arduino UNO:

The Arduino Uno is a popular microcontroller board that operates based on the ATmega328P microcontroller. Here's a basic overview of how the Arduino Uno operates:

Microcontroller: The heart of the Arduino Uno is the ATmega328P microcontroller. It is responsible for executing the code loaded onto the board and controlling the various inputs and outputs.

Clock Speed: The ATmega328P on the Arduino Uno typically operates at 16 MHz, providing the timing for the execution of instructions in the code.

Power Supply: The Arduino Uno can be powered in several ways. It can be powered through the USB connection to a computer, an external power supply, or a battery. The onboard voltage regulator ensures a stable 5V supply for the board's components.

Programming: Users write programs, or sketches, using the Arduino IDE (Integrated Development Environment) on a computer.

USB Interface:

The Arduino Uno has a built-in USB-to-Serial converter that allows it to communicate with a computer. This USB interface is used for both programming the board and for serial communication between the Arduino and a computer.

Input/Output Pins: The Arduino Uno has a set of digital and analog input/output pins that can be used to connect sensors, actuators, and other electronic components. Digital pins can be configured as either input or output, while analog pins can read analog signals.

PWM (Pulse Width Modulation): Some of the digital pins on the Arduino Uno support PWM, allowing for analog-like output control. PWM is commonly used for tasks such as dimming LEDs or controlling the speed of motors.

Built-in LED: The Arduino Uno has a built-in LED on pin 13, which is often used for basic testing and debugging. This LED is connected to the microcontroller and can be controlled through software.

Reset Button: The reset button on the Arduino Uno restarts the microcontroller, allowing users to re-upload code without disconnecting the board.

6.6 AT-MEGA16 MICROCONTROLLER:

The Atmega16 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture. By executing powerful instructions in a single clock cycle, the Atmega16 achieves throughputs approaching 1 MIPS per MHz allowing the system designed to optimize power consumption versus processing speed.

The AVR core combines a rich instruction set with 32 general purpose working registers. All the 32 registers are directly connected to the Arithmetic Logic Unit (ALU), allowing two independent registers to be accessed in one single instruction executed in one clock cycle. The resulting architecture is more code efficient while achieving throughputs up to ten times faster than conventional CISC microcontrollers.

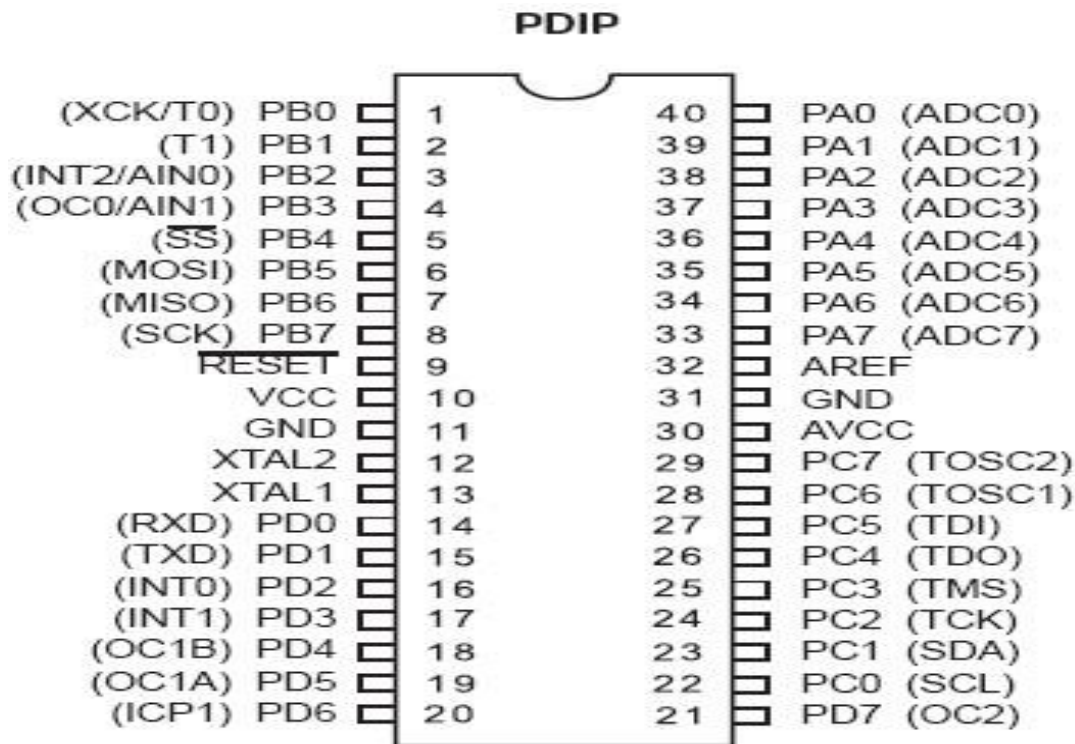
The Atmega16 provides the following features: 8K bytes of In-System Programmable Flash Program memory with Read-While-Write capabilities, 1024 bytes EEPROM, 2K byte SRAM, 32 general purpose I/O lines, 32 general purpose working registers, a JTAG interface for Boundary scan, On-chip Debugging support and programming, three flexible Timer/Counters with compare modes, Internal and External Interrupts, a serial programmable USART, a byte oriented Two-wire Serial Interface, an 8-channel, 10-bit ADC with optional differential input stage with programmable gain (TQFP package only), a programmable Watchdog Timer with Internal Oscillator, an SPI serial port, and six software selectable power saving modes. The Idle mode stops the CPU while allowing the USART, Two-wire interface, A/D Converter, SRAM, Timer/Counters, SPI port, and interrupt system to continue functioning. The Power-down mode saves the register contents but freezes the Oscillator, disabling all other chip functions until the next External Interrupt or Hardware Reset. In Power-save mode, the Asynchronous Timer continues to run, allowing the user to maintain a timer base while the rest of the device is sleeping. The ADC Noise Reduction mode stops the CPU and all I/O modules except Asynchronous Timer and ADC, to minimize switching noise during ADC conversions. In Standby mode, the crystal/resonator Oscillator is running while the rest of the device is sleeping. This allows very fast start-up combined with low-power consumption. In Extended Standby mode, both the main Oscillator and the Asynchronous Timer continue to run.

The device is manufactured using Atmel's high density nonvolatile memory technology. The On chip ISP Flash allows the program memory to be reprogrammed in-system through an SPI serial interface, by a conventional nonvolatile memory programmer, or by an On-chip Boot program running on the AVR core. The boot program can use any interface to download the

application program in the Application Flash memory. Software in the Boot Flash section will continue to run while the Application Flash section is updated, providing true Read-While-Write operation. By combining an 8-bit RISC CPU with In-System Self-Programmable Flash on a monolithic chip, the Atmel ATmega16 is a powerful microcontroller that provides a highly-flexible and cost-effective solution to many embedded control applications.

FEATURES

- **High-performance, low-power AVR & 8-bit Microcontroller.**
- **Advanced RISC Architecture**
 - 131 Powerful Instructions – Most Single-clock Cycle Execution
 - 32 x 8 General Purpose Working Registers
 - Fully Static Operation
 - Up to 16 MIPS Throughput at 16 MHz
 - On-chip 2-cycle Multiplier
- **Operating Voltages**
 - 2.7 - 5.5V for ATmega16L
 - 4.5 - 5.5V for ATmega16
- **Speed Grades**
 - 0 - 8 MHz for ATmega16L
 - 0 - 16 MHz for ATmega16
- **Power Consumption at 1 MHz, 3V, 25°C for ATmega16L**
 - Active: 1.1 mA
 - Idle Mode: 0.35 mA
 - Power-down Mode: < 1 Ma

PIN CONFIGURATION:**Figure 6.7** Pin Diagram of ATMEGA16**Pin Descriptions:**

VCC: Digital supply voltage.

GND: Ground.

Port A (PA7..PA0): Port A serves as the analog inputs to the A/D Converter.

Port A also serves as an 8-bit bi-directional I/O port, if the A/D Converter is not used. Port pins can provide internal pull-up resistors (selected for each bit). The Port A output buffers have symmetrical drive characteristics with both high sink and source capability. When pins PA0 to PA7 are used as inputs and are externally pulled low, they will source current if the internal pullup resistors are activated. The Port A pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Port B (PB7..PB0): Port B is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port B output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port B pins that are externally pulled low will source current if the pull-up resistors are activated. The Port B pins are tri-stated when a reset condition becomes active, even if the clock is not running.

Port C (PC7..PC0): Port C is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port C output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port C pins that are externally pulled low will source current if the pull-up resistors are activated. The Port C pins are tri-stated when a reset condition becomes active, even if the clock is not running. If the JTAG interface is enabled, the pull-up resistors on pins PC5(TDI), PC3(TMS) and PC2(TCK) will be activated even if a reset occurs. The TD0 pin is tri-stated unless TAP states that shift out data are entered.

Port D (PD7..PD0): Port D is an 8-bit bi-directional I/O port with internal pull-up resistors (selected for each bit). The Port D output buffers have symmetrical drive characteristics with both high sink and source capability. As inputs, Port D pins that are externally pulled low will source current if the pull-up resistors are activated. The Port D pins are tri-stated when a reset condition becomes active, even if the clock is not running.

RESET: Reset Input. A low level on this pin for longer than the minimum pulse length will generate a reset, even if the clock is not running.

XTAL1: Input to the inverting Oscillator amplifier and input to the internal clock operating circuit. **XTAL2:** Output from the inverting Oscillator amplifier.

AVCC: AVCC is the supply voltage pin for Port A and the A/D Converter. It should be externally connected to VCC, even if the ADC is not used. If the ADC is used, it should be connected to VCC through a low-pass filter.

AREF: AREF is the analog reference pin for the A/D Converter.

6.7 POWER SUPPLY DESCRIPTION

A variable regulated power supply, also called a variable bench power supply, is one where you can continuously adjust the output voltage to your requirements. Varying the output of the power supply is the recommended way to test a project after having double checked parts placement against circuit drawings and the parts placement guide. This type of regulation is ideal for having a simple variable bench power supply. Actually this is quite important because one of the first projects a hobbyist should undertake is the construction of a variable regulated power supply. While a dedicated supply is quite handy e.g. 5V or 12V, it's much handier to have a variable supply on hand, especially for testing. Most digital logic circuits and processors need a 5 volt power supply. To use these parts we need to build a regulated 5 volt source. Usually you start with an unregulated power supply ranging from 9 volts to 24 volts DC (A 12 volt power supply is included with the Beginner Kit and the Microcontroller Beginner Kit). To make a 5 volt power supply, we use a LM7805 voltage regulator IC (Integrated Circuit). The IC is shown below.

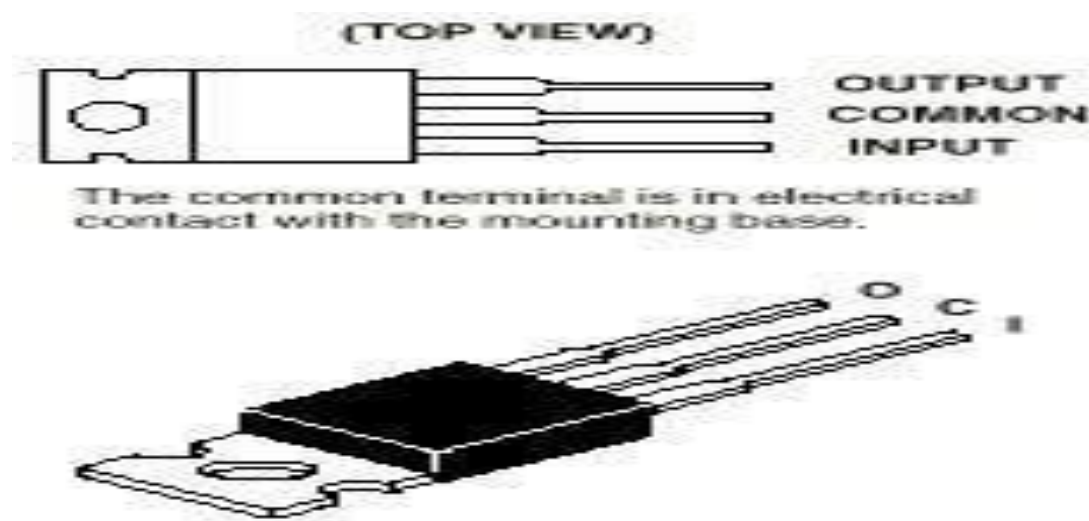


Figure 6.8: LM7805

The LM7805 is simple to use. You simply connect the positive lead of your unregulated DC power supply (anything from 9VDC to 24VDC) to the Input pin, connect the negative lead to the Common pin and then when you turn on the power, you get a 5 volt supply from the Output pin.

CHAPTER 7

SOFTWARE CONSTRAINTS

7.1 SOURCE CODE

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(8,9,10,11,12,13);//rs,en,data pins d4 -d7


unsigned int MQ7;////A0
unsigned int MQ135;/////A1
unsigned int MQ4;/////A2


int buzz=7;


void setup()
{
  Serial.begin(9600);
  lcd.begin(16,2);
  pinMode(buzz, OUTPUT);

  //////////////////////////////////////

  lcd.clear();
  lcd.setCursor(0,0);lcd.print("INDUSTRIAL ");
  lcd.setCursor(0,1);lcd.print("POLLUTION ");
  delay (3000);
  //////////////////////////////////////

  lcd.clear();
  lcd.setCursor(0,0);lcd.print("MONITORING USING");
  lcd.setCursor(0,1);lcd.print("LABVIEW AND ARDUINO");
  delay (3000);
  lcd.clear();
  //lcd.setCursor(0,0);lcd.print("C:  N:  G:  ");
```

```

}

void loop()
{

    digitalWrite(buzz, LOW);
    MQ7 = analogRead(0);MQ7=MQ7/4;
    lcd.setCursor(0,0);lcd.print("CO: ");lcd.setCursor(5,0);lcd.print(MQ7);
    Serial.print("C:");
    if((MQ7 >= 0) & (MQ7 <= 9)){Serial.print("00");Serial.print(MQ7);}
    if((MQ7>= 10) & (MQ7 <= 99)){Serial.print("0");Serial.print(MQ7);}
    if((MQ7 >= 99) & (MQ7 <= 999)){Serial.print(MQ7);}
    if(MQ7>150)
    {
        digitalWrite(buzz, HIGH);delay(3000);
    }
    delay(1000);
    //////////////////////////////////////
    MQ135 = analogRead(1);MQ135=MQ135/4;
    lcd.setCursor(9,0);lcd.print("N:");lcd.setCursor(12,0);lcd.print(MQ135);
    Serial.print("N:");
    if((MQ135 >= 0) & (MQ135 <= 9)){Serial.print("00");Serial.print(MQ135);}
    if((MQ135>= 10) & (MQ135 <= 99)){Serial.print("0");Serial.print(MQ135);}
    if((MQ135 >= 99) & (MQ135 <= 999)){Serial.print(MQ135);}
    if(MQ135>150)
    {
        digitalWrite(buzz, HIGH);delay(3000);
    }
    delay(1000);
    //////////////////////////////////////

```

```
MQ4 = analogRead(2);MQ4=MQ4/4;
lcd.setCursor(0,1);lcd.print("G:");lcd.setCursor(3,1);lcd.print(MQ4);
Serial.print("G:");
if((MQ4 >= 0) & (MQ4 <= 9)){Serial.print("00");Serial.print(MQ4);}
if((MQ4>= 10) & (MQ4 <= 99)){Serial.print("0");Serial.print(MQ4);}
if((MQ4 >= 99) & (MQ4 <= 999)){Serial.print(MQ4);}
if(MQ4>150)
{
digitalWrite(buzz, HIGH);delay(3000);
}
delay(1000);
lcd.clear();
}
```

CHAPTER 8

PROJECT DESCRIPTION

8.1 OVERVIEW:

The "Industrial Pollution Monitoring using LabVIEW and Arduino" project addresses the critical need for effective and real-time monitoring of industrial emissions. This innovative solution integrates LabVIEW, a graphical programming platform, with Arduino, a versatile and cost-effective microcontroller, to create a comprehensive system for measuring and analyzing pollutants in industrial settings.

8.2 OBJECTIVES:

Real-time Monitoring: Develop a system that provides real-time data on key pollutants emitted by industrial processes.

Efficient Data Acquisition: Implement Arduino-based sensor modules for precise and continuous measurement of particulate matter, gases, and environmental parameters.

Seamless Communication: Establish seamless communication between Arduino and LabVIEW to facilitate efficient data transfer and analysis.

User-friendly Interface: Create a user-friendly interface on LabVIEW for easy data visualization, trend analysis, and alarm notifications based on predefined thresholds.

Cost-effectiveness and Scalability: Utilize the cost-effective Arduino platform to ensure the economic viability and scalability of the monitoring solution for widespread industrial use.

Methodology: The project involves the deployment of Arduino-based sensor modules strategically placed within an industrial environment. These sensors continuously collect data on pollutants, including particulate matter and gases. The collected data is processed by the Arduino microcontroller, which calibrates the measurements for accuracy and reliability.

LabVIEW, with its intuitive graphical interface, establishes communication with Arduino to acquire the processed data. The LabVIEW interface allows for real-time monitoring, providing visual representations of pollutant levels and environmental conditions. Additionally, the system enables trend analysis and issues instant notifications when pollutant levels exceed predefined thresholds.

8.3 SIGNIFICANCE:

This project holds immense significance in revolutionizing the way industries monitor and manage pollution. By combining the strengths of LabVIEW and Arduino, the solution offers a cost-effective, scalable, and user-friendly approach to industrial pollution monitoring. It empowers industries to make informed decisions, meet regulatory standards, and contribute to sustainable environmental practices.

8.4 EXPECTED OUTCOMES:

Accurate and Timely Data: The system will provide accurate and timely data on industrial pollutants, aiding in quick decision-making.

Improved Environmental Management: Industries can use the collected data for better environmental management and compliance with regulations.

Cost-effective Solution: The project aims to demonstrate a cost-effective solution, ensuring accessibility for industries with varying budget constraints.

Scalability for Various Industries: The modular design allows for scalability, making the system adaptable to the diverse needs of different industrial sectors.

8.5 DATA FLOW DIAGRAM IN LABVIEW:

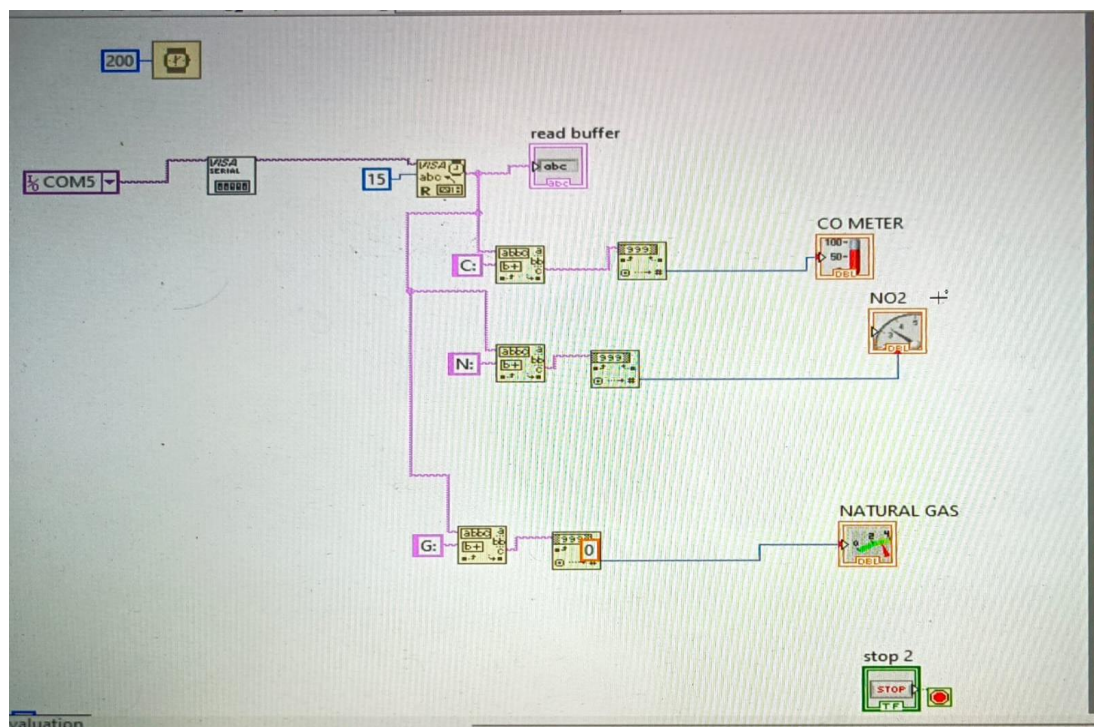


Figure 8.1: Data Flow Diagram in LabVIEW

The Arduino sensors, such as gas detectors and particulate matter sensors, capture real-time data, which is then processed by LabVIEW to generate meaningful insights. LabVIEW's user-friendly interface allows for the creation of customizable dashboards and graphical representations of pollution levels, aiding in quick decision-making for industrial operators. The system can be designed to trigger alerts or alarms when pollution levels surpass predefined thresholds, enabling prompt corrective actions.

The synergy between LabVIEW and Arduino enhances the efficiency of industrial pollution monitoring by providing a comprehensive and accessible platform for data analysis. This integration facilitates the continuous monitoring of environmental parameters, contributing to sustainable industrial practices and the reduction of adverse impacts on ecosystems and public health. Overall, the combination of LabVIEW and Arduino offers a robust solution for real-time industrial pollution management and control.

WORKING PROTOTYPE:

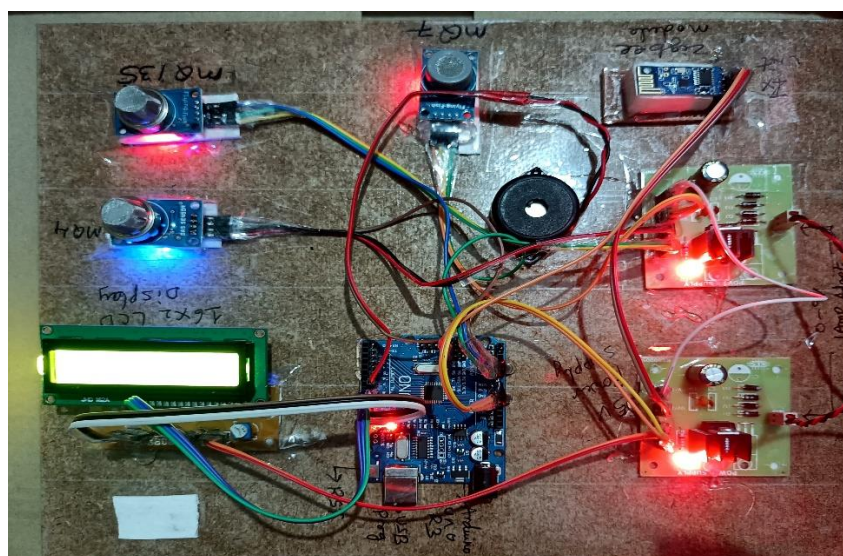


Figure 8.2: Working Prototype

CHAPTER 9

RESULT

The implementation of the Industrial Pollution Monitoring system using LabVIEW and Arduino yields significant outcomes for environmental management. Real-time data acquisition and analysis enable timely identification of pollutant spikes, empowering industrial operators to take immediate corrective actions. LabVIEW's graphical interface provides a user-friendly platform for visualizing pollution trends, facilitating comprehensive decision-making.

By utilizing Arduino sensors for data collection, the project ensures accurate and reliable measurements of various pollutants, including gases and particulate matter. The system's ability to generate alerts upon surpassing predefined thresholds enhances responsiveness, enabling proactive measures to mitigate environmental impact. The project's results include improved regulatory compliance, reduced emissions, and enhanced overall environmental sustainability in industrial operations.

Moreover, the accessible and customizable dashboards created through LabVIEW simplify data interpretation, aiding in informed decision-making. The combination of LabVIEW and Arduino not only streamlines pollution monitoring but also contributes to long-term environmental stewardship by fostering a culture of responsible industrial practices. Ultimately, the project's success lies in its ability to integrate seamlessly into industrial processes, providing a practical and efficient solution for ongoing pollution management and control.

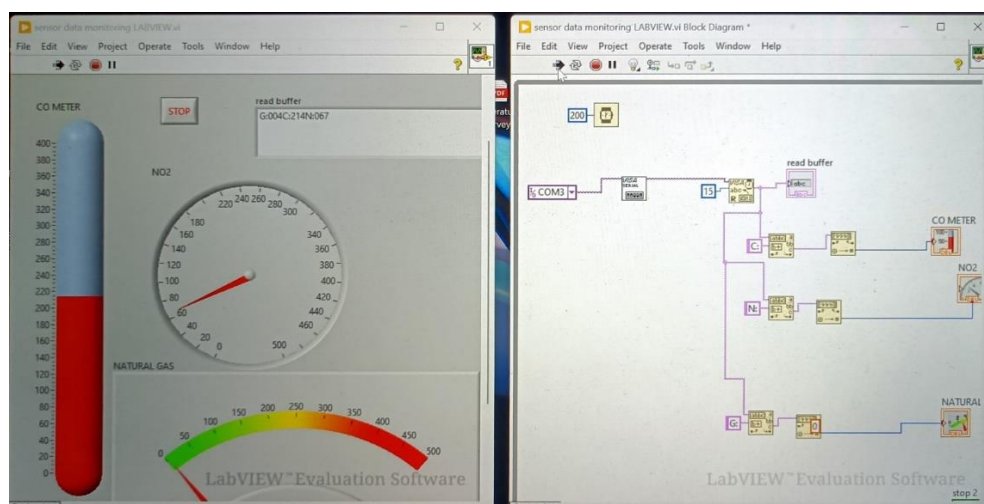


Figure 9.1: Result

CHAPTER 10

ADVANTAGES AND DISADVANTAGES

ADVANTAGES:

- LabVIEW can handle data from multiple sensors and manage it effectively.
- Arduino's wide range of compatible sensors allows you to monitor various pollutants specific to your industry.
- Arduino is a relatively low-cost microcontroller platform, LabVIEW offers free open-source versions for hobbyists and students, reducing software licensing costs.

DISADVANTAGES:

- Dependency on Technology
- User Acceptance

CHAPTER 11

CONCLUSION

In conclusion, the Industrial Pollution Monitoring project employing LabVIEW and Arduino stands as a commendable stride towards sustainable industrial practices and environmental stewardship. The integration of these technologies has resulted in a robust system that seamlessly acquires, processes, and visualizes real-time data on various pollutants. This not only enhances the efficiency of industrial operations but also contributes significantly to regulatory compliance and public health.

The project's ability to promptly identify and respond to pollution spikes empowers industries to proactively address environmental concerns, minimizing the ecological footprint. The synergy between LabVIEW's user-friendly interface and Arduino's precise sensors ensures accurate data collection, fostering reliable insights into pollutant levels. The customizable dashboards generated by LabVIEW simplify data interpretation, enabling stakeholders to make informed decisions swiftly.

Beyond immediate benefits, the project establishes a foundation for a culture of responsible industrial practices, promoting long-term environmental sustainability. By reducing emissions and facilitating continuous monitoring, it aligns with global efforts to combat climate change and environmental degradation. In essence, the Industrial Pollution Monitoring project emerges as a vital tool, not just for regulatory compliance but as a catalyst for positive change, harmonizing industrial processes with ecological well-being.

CHAPTER 12

FUTURE SCOPE

The future scope of an "industrial pollution monitoring using LabVIEW and Arduino" project is promising. With increasing environmental concerns, there's a growing demand for efficient and real-time pollution monitoring systems. This project can be expanded to incorporate advanced sensors, enhancing accuracy and scope. Integrating IoT technologies can enable remote monitoring and data analytics, providing valuable insights for industrial sustainability. Collaboration with regulatory bodies and industries can lead to widespread implementation. The project's adaptability to emerging pollutants and adherence to evolving environmental standards make it a vital tool for ensuring cleaner industrial practices. Future developments may involve machine learning algorithms for predictive analysis and anomaly detection. Scaling the project to encompass smart cities or industrial zones contributes to comprehensive environmental management. Continuous updates to LabVIEW and Arduino platforms offer opportunities for refining and expanding the project's capabilities. Incorporating community engagement features could raise awareness and involve citizens in pollution control initiatives. Overall, the project has the potential to play a crucial role in fostering environmental responsibility and sustainable industrial practices in the years to come.

CHAPTER 13

REFERENCES:

1. A.R. Al-Ali, Imran Zulkarnaen and Fadi Aloul, “A Mobile GPRS-Sensors Array for Air Pollution Monitoring” IEEE SENSORS JOURNAL, VOL. 10, NO. 10, OCTOBER 2010
2. Shashikant U.Suryawanshi, Deep Ganga Dhang ,Ashish A. Chou Gule Shailendra B. Mote , “Implementation Of Embedded Wireless Air Pollution Monitoring System” IOSR Journal of Electronics and Communication Engineering (IOSR-JMCE) ISSN: 2278-2834-, ISBN: `2278-8735, PP: 27-30.
3. Aruljothi.R , “Air Pollution Measuring System with Mobile Sensor Arrays” International Journal of Scientific & Engineering Research, Volume 4, Issue 5, May-2013 ISSN 2229-5518
4. Rakesh Kumar Giri, “An Itinerant GPRS-GPS and Sensors Integration for Atmospheric Effluence Screening” International Journal of Technology and Engineering System (IJTES): Jan –March 2011- Vol.2. No.2.
5. N. Kula Ratna and B. H. Susantha, “An environmental air pollution monitoring system based on the IEEE 1451 standard for low cost requirements,”IEEE Sensors J., vol. 8, pp. 415–422, Apr. 2008.
6. F. Tsou, E Forzani, A. Rai, R. Wang, R. Tsui, S. Mastroianni, C. Knobbe, A. J. Gandolfi, and N. J. Tao, “A wearable and wireless sensor system for real-time monitoring of toxic environmental volatile organic compounds,” IEEE Sensors J., vol. 9, pp. 1734–1740, Dec.2009.
7. Y. J. Jung, Y. K. Lee, D. G. Lee, K. H. Ryu, and S. Mittel, “Air pollution monitoring system based on Grosvenor network,” in Proc. IEEE Int. Geoscience Remote Sensing Symp., 2008, vol. 3, pp. 1370–1373.
8. C. J. Wong, M. Z. MatJafri, K. Abdullah, H. S. Lim, and K. L. Low, “Temporal air quality monitoring using surveillance camera,” in Proc. IEEE Int. Geoscience and Remote Sensing Symp., 2007, pp.2864–2868.
9. M. Gao, F. Zhang, and J. Tian, “Environmental monitoring system with wireless mesh network based on embedded system,” in Proc. 5th IEEE Int. Symp. Embedded Computing, 2008, pp. 174–179.

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